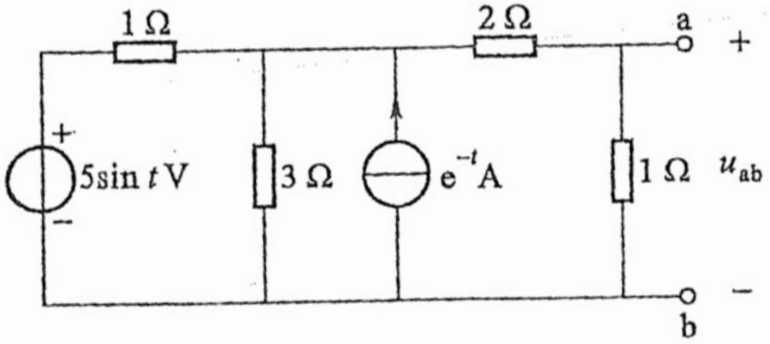
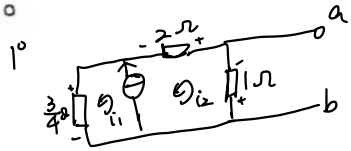
4-1 应用叠加定理求题 4-1 图所示电路中电压 u_{ab}。



题 4-1 图

1° → 是断路



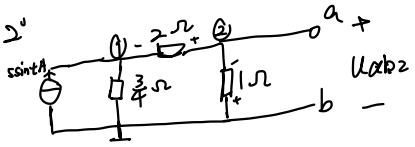
直接用电阻与流算 1Ω 上的电流

$\frac{3}{4}i_1 + 1\times i_2 + 2i_2 = 0$

$i_1 - i_2 = e^{-t}A$

$U_{ab1} + 1\times i_2 = 0 \Rightarrow U_{ab1} = \frac{1}{5}e^{-t}V$

2°



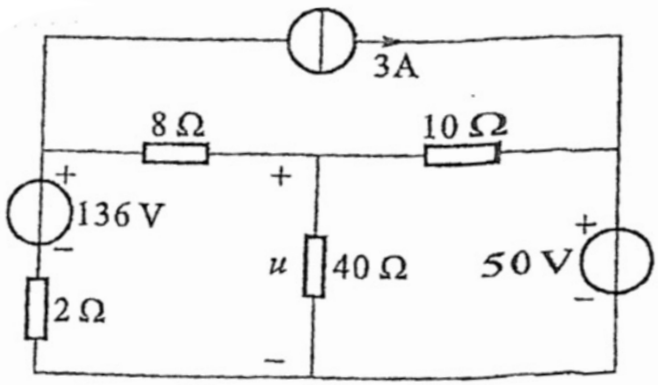
$(\frac{4}{3}+\frac{1}{2})U_1 - \frac{1}{2}U_2 = 5\sin t$

$(1+\frac{1}{2})U_2 - \frac{1}{2}U_1 = 0$ $\Rightarrow U_{ab2} = \sin t\,V$

$U_2 = U_{ab2}$

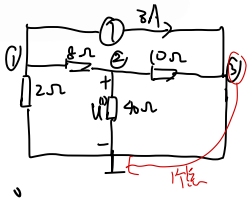
$\Rightarrow U_{ab} = U_{ab1} + U_{ab2} = \frac{1}{5}e^{-t} + \sin t\ (V)$

4-2 应用叠加定理求题 4-2 图所示电路中电压 u。



题 4-2 图

1°



$(\frac{1}{2}+\frac{1}{8})U_1 - \frac{1}{8}U_2 = -3A$

$(\frac{1}{40}+\frac{1}{8}+\frac{1}{10})U_2 - \frac{1}{8}U_1 - \frac{1}{10}U_3 = 0$

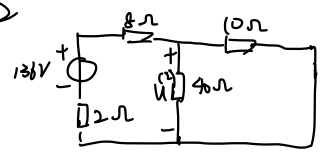
$\frac{1}{10}U_3 - \frac{1}{10}U_2 = 3A$

$U_2 = U^{(1)}$

$\Rightarrow U_1 = 0$, $U_2 = 24V$, $U_3 = 54V$

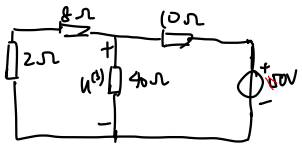
$\Rightarrow U^{(1)} = 24V$

2°



$U^{(2)} = \frac{\frac{1}{40}+\frac{1}{10}}{8+2+\frac{1}{\frac{1}{40}+\frac{1}{10}}} \times 136 = \frac{544}{9}V$

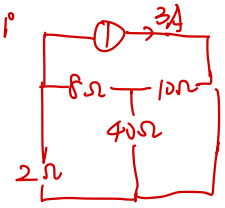
3°



$U^{(3)} = \frac{\frac{1}{40}+\frac{1}{10}}{10+\frac{1}{\frac{1}{40}+\frac{1}{10}}} \times 150 = \frac{200}{9}V$

$U = U^{(1)} + U^{(2)} + U^{(3)} = 151.11V$ ✗

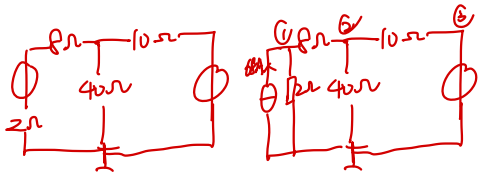
分成2个：1°



电流方流 $\Rightarrow i = \frac{1}{15}A$

$\Rightarrow U^{(1)} = -40\times\frac{1}{15} = -\frac{8}{3}V$

2°



$U_3 = 30V$

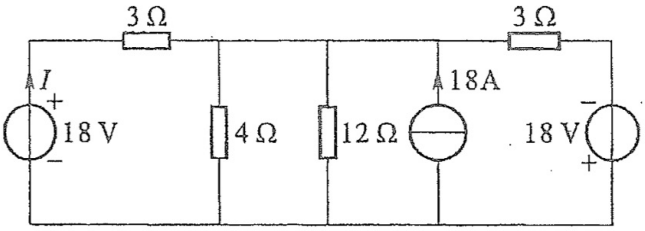
$(\frac{1}{2}+\frac{1}{8})U_1 - \frac{1}{8}U_2 = 6\frac{3}{8}$

$(\frac{1}{8}+\frac{1}{40}+\frac{1}{10})U_2 - \frac{1}{8}U_1 - \frac{1}{10}U_3 = 0$

$\Rightarrow U_2 = \frac{248}{3}V = U^{(2)}$

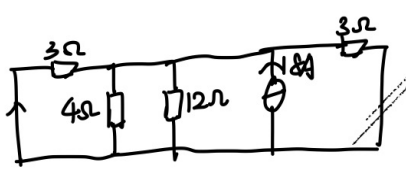
$U = U^{(1)} + U^{(2)} = 80V$

4-3 应用叠加定理求题 4-3 图所示电路中的电流 I。



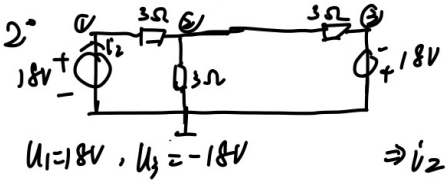
题 4-3 图

1°



$i_1 = -18\times\frac{4//12//3}{3+4//12//3} = -6A$

2°



$U_1 = 18V$, $U_3 = -18V$

$-\frac{1}{3}U_1 + (\frac{1}{3}+\frac{1}{3}+\frac{1}{3})U_2 - \frac{1}{3}U_3 = 0$

$\Rightarrow U_2 = 0$

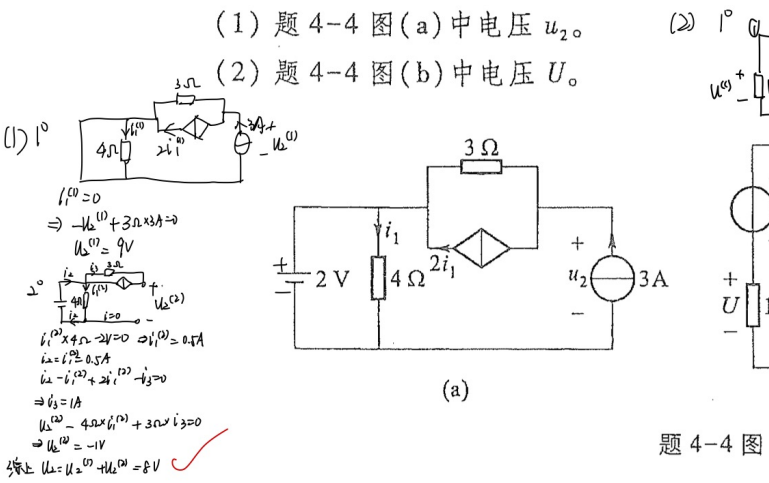
$\Rightarrow i_2 = \frac{U_1 - U_2}{3\Omega} = 6A$

$I = i_1 + i_2 = 0$

4-4 应用叠加定理时，将受控源都保留在分电路中。求：

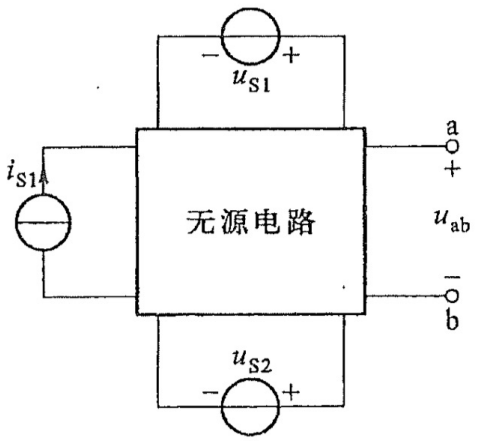
(1) 题 4-4 图(a)中电压 u_2。

(2) 题 4-4 图(b)中电压 U。



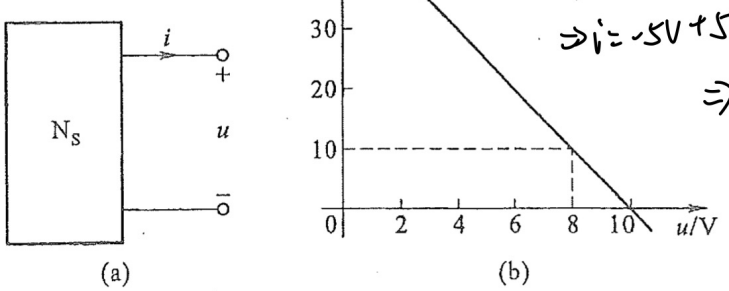
题 4-4 图

4-7 题 4-7 图所示电路中，当电流源 i_{S1} 和电压源 u_{S1} 反向时(u_{S2} 不变)，电压 u_{ab} 是原来的 0.5 倍；当 i_{S1} 和 u_{S2} 反向时(u_{S1} 不变)，电压 u_{ab} 是原来的 0.3 倍。仅 i_{S1} 反向(u_{S1}、u_{S2} 均不变)时，电压 u_{ab} 应为原来的几倍？



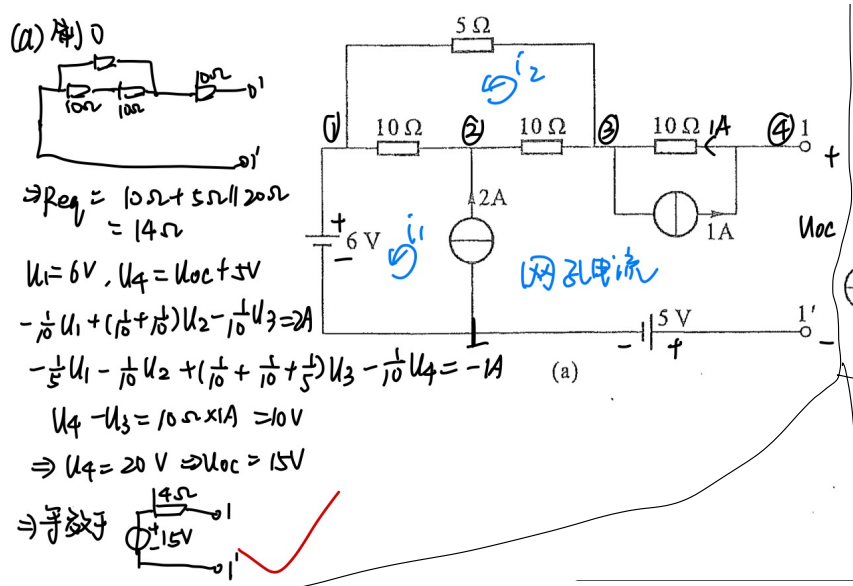
题 4-7 图

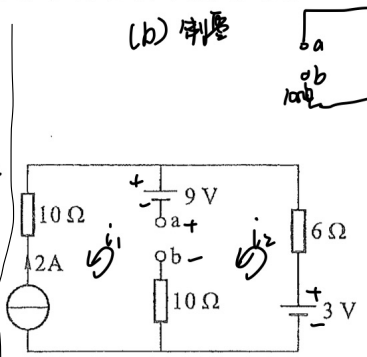
4-11 题 4-11 图(a)所示含源一端口的外特性曲线画于题 4-11 图(b)中，求其等效电源。

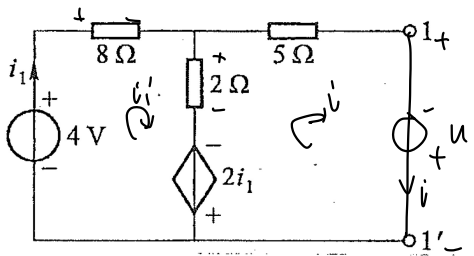


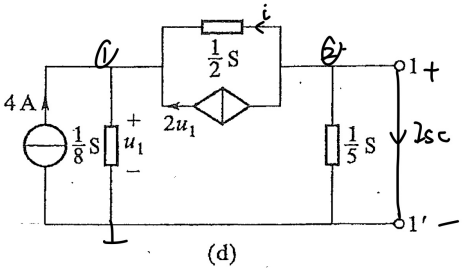
题 4-11 图

4-12 求题 4-12 图所示各电路的戴维南等效电路或诺顿等效电路。

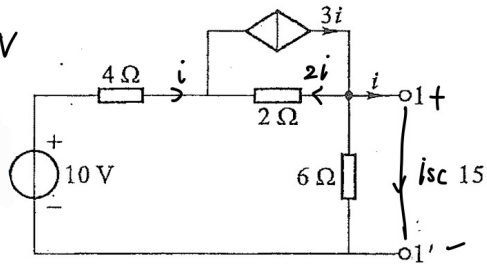


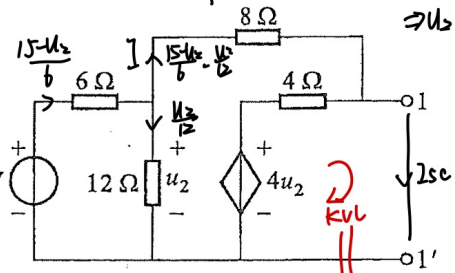




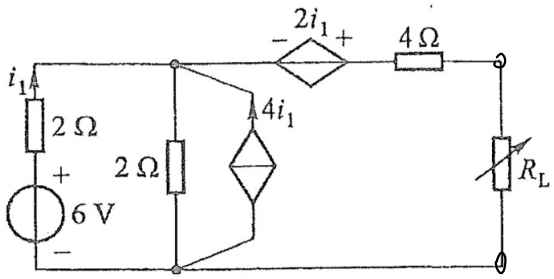


4-13 求题 4-13 图所示两个一端口的戴维南等效电路或诺顿等效电路，并解释所得结果。





4-17 题 4-17 图所示电路的负载电阻 R_L 可变，R_L 等于何值时可吸收最大功率？求此功率。



题 4-17 图